

National Institute of Technology, Kurukshetra

BTech (IT/M&C), End Term Exam (Theory), July/August'2026

Course no.: ITPC-206 (IT)/CSPC-206 (M&C)
Course name: Database Management Systems
No. of Questions to be attempted: 05
No. of Pages used: 02

Date of Exam: 07 August, 2026
Time allowed: 03 Hrs
Maximum marks: 50
Roll no.: _____

Note: Answer all the questions. You may answer the questions in any order. However, all parts of the same question must be answered together. Clearly state any reasonable assumption you make.

Q.1 Answer the following (briefly):

- "Global University" records every student's roll number, attendance percentage, and semester exam scores as raw data in its academic system. The university's reporting engine processes this data to generate student progress reports for professors. These reports identify students with low attendance, highlight those failing multiple subjects, and predict students at risk of academic probation. Identify two examples of data and two examples of information from the above.
- Consider a table *Professor* (ProfID, Name, Projects), where Projects is a multi-valued attribute listing research project names. List three specific anomalies that can occur.
- Data retrieval is different from information retrieval. Explain briefly, with suitable example.
- Why SQL does not allow to compare an attribute value to NULL using '=' and '< >' operators? How this problem is resolved in SQL?
- A transaction fails after entering the 'Partially Committed' state but before its final commit. Which ACID property is not upheld?

Q.2 (a) A pharmacy chain needs to track prescriptions, drugs, and interactions. Each drug has a unique DIN, name, and manufacturer. Doctors (with license number and specialty) prescribe drugs to patients. A prescription has a date, dosage, and refill information. Critically, the system must track drug interactions - certain drugs cannot be taken together. Pharmacists must verify that new prescriptions don't conflict with a patient's current medications. Draw an ERD for the above data requirements, with appropriate entities (strong & weak), attributes (PK, derived, multi, & complex), relationships (recursive, etc) and cardinalities (Min & Max).

(b) The Ministry of Health is implementing a centralized DB for the AB-PMJAY to streamline patient management across empaneled hospitals. The core entity is the Patient, who possesses a unique PatientID and a general MedicalHistory. However, patients access treatment through different financial channels. A significant number are AB-PMJAY Beneficiary patients, identified by a FamilyID and a predefined EntitlementAmount. Other patients are covered under various StateHealthScheme Beneficiary programs, which require a SchemeCardNo and offer StateSpecificBenefits. Some patients are PrivateInsurance Holders, with attributes like PolicyNo and their Insurer company. Finally, there are CashPatients who pay directly, for whom the PaymentMode and any applicable Discount must be recorded. Each of these patient categories involves a distinctly different approval process and is subject to different treatment package rates. Develop an Enhanced ERD to model this system. Your diagram must clearly illustrate the use of specialization/generalization to represent the different types of patients.

Q.3 Consider the Company relational DB, where the primary key(s) are underlined.

employee (employee-name, gender, salary, city, country)
company (company-name, city, country)
works (employee-name, company-name)
manages (employee-name, manager-name)

(a) Give an equivalent SQL expression for each of the following data retrieval queries:

- Find employees whose name starts with 'Dr.' or 'Prof.' but are not assigned to any company.
- Write an SQL query to find the names of employees who work in the same city as their

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company.

- (iii) Find the average salary for male & female employees within each company. Only show results where the difference between male & female average salary within the same company is greater than 10000.
- (iv) Find employees who are NOT managers AND work for a company that has at least one manager as an employee.
- (b) Give an equivalent SQL expression or answer to each of the following queries:
- (i) How would you remove the table if you later need to recover the data structure for a backup? (6)
- (ii) What referential action (e.g., CASCADE, SET NULL) would be appropriate for the foreign key from works.employee-name to employee.employee-name if an employee is deleted? Justify your answer.
- (iii) Truncate Table Company, Will this succeed? Why or why not?
- (iv) Write an SQL statement to delete all employees who work for a company headquartered in a city where there are no other companies.
- (a) What is a functional dependency? What are the possible sources of the information that defines the functional dependencies that hold among the attributes of a relation schema? (2)
- (b) For order processing: $R(\text{OrderID}, \text{ProductID}, \text{CustomerID}, \text{Warehouse}, \text{City}, \text{Quantity})$ with FDs: (i) $\text{OrderID}, \text{ProductID} \rightarrow \text{Quantity}$, $\text{CustomerID} \rightarrow \text{City}$ (ii) $\text{CustomerID} \rightarrow \text{City}$. (iii) $\text{Warehouse} \rightarrow \text{City}$. Prove inference of: (3)
- (i) $\text{OrderID}, \text{ProductID} \rightarrow \text{City}$
- (ii) Warehouse cannot supply to multiple cities.
- (c) Consider the Order system: $R(\text{OrderID}, \text{CustomerID}, \text{ProductID}, \text{Quantity}, \text{Price}, \text{Total})$ with FDs: (FD1) $\text{OrderID} \rightarrow \text{CustomerID}$, OrderID , (FD2) $\text{ProductID} \rightarrow \text{Quantity}, \text{Price}$, and (FD3) $\text{Price}, \text{Quantity} \rightarrow \text{Total}$. For the decomposition: $R_1(\text{OrderID}, \text{CustomerID})$, $R_2(\text{OrderID}, \text{ProductID}, \text{Quantity}, \text{Price})$ and $R_3(\text{Price}, \text{Quantity}, \text{Total})$, verify both properties (Lossless, dependency preserving). (3)
- (d) Given $R(A,B,C,D,E)$ with FDs: $\{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C, AC \rightarrow D\}$. Find candidate keys, identify the current norm form (NF) and decompose the relation to its highest NF. (4)
- (a) What is the system log used for? What are the typical kinds of records in a system log? What are transaction commit points, and why are they important? (2)
- (b) You are transferring ₹10,000 from your Savings Account (S) to your Brokerage Account (B) via a banking portal. The database executes these two SQL queries sequentially as a single transaction: (4)
- Query 1:** `UPDATE Savings SET Balance = Balance - 10000 WHERE AccNo = 'S';`
- Query 2:** `UPDATE Brokerage SET Balance = Balance + 10000 WHERE AccNo = 'B';`
- Query 1 executes successfully. Your Savings balance decreases. However, just as Query 2 begins execution, the database server experiences a segmentation fault and crashes.
- (i) Which ACID property is violated when the server comes back online? Explain why.
- (ii) What is the state of your total net worth across both accounts after the crash? (Calculate the exact figures).
- (c) Consider the three DB transactions T_1, T_2 , and T_3 and a schedule $S: r_1(X), r_2(Z), r_1(Z), r_3(X), r_3(Y), w_1(X), w_3(Y), r_2(Y), w_2(Z), w_2(Y)$. The actions are listed in the order they are scheduled and prefixed with the transaction name. Evaluate and discuss whether Schedule (s) is serializable or not? (4)